

COMPLEXITY AND TIME

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Abstract

A large literature shows that people's valuation of delayed financial rewards violates exponential discounting, exhibiting a hyperbolic pattern: high short-run impatience that strongly decreases in the length of the delay. We test the hypothesis that the hyperbolic pattern in measured discount rates over money reflects mistakes driven by the complexity of evaluating delayed payoffs. We document that hyperbolicity (i) is strongly associated with choice inconsistency and cognitive uncertainty, (ii) increases in overt complexity manipulations, and (iii) arises nearly identically in computationally similar tasks that involve no actual payoff delays. Our results suggest that even if people had exponential discount functions, complexity-driven mistakes would cause them to make hyperbolic choices. We examine which experimental techniques to estimate present bias are (not) confounded by information-processing constraints. (JEL: C91, D91, G0)

1. Introduction

Since the seminal work of Thaler (1981), behavioral economists have gathered significant evidence that people do not discount future financial flows exponentially as suggested by standard economic models (see Cohen et al. 2020; Ericson and Laibson 2019 for reviews). The typical experiments use money-early-versus-later designs, which elicit the present value of monetary payouts to be made at later dates. Figure 1 shows typical results using our own data, plotting delayed payments on the horizontal

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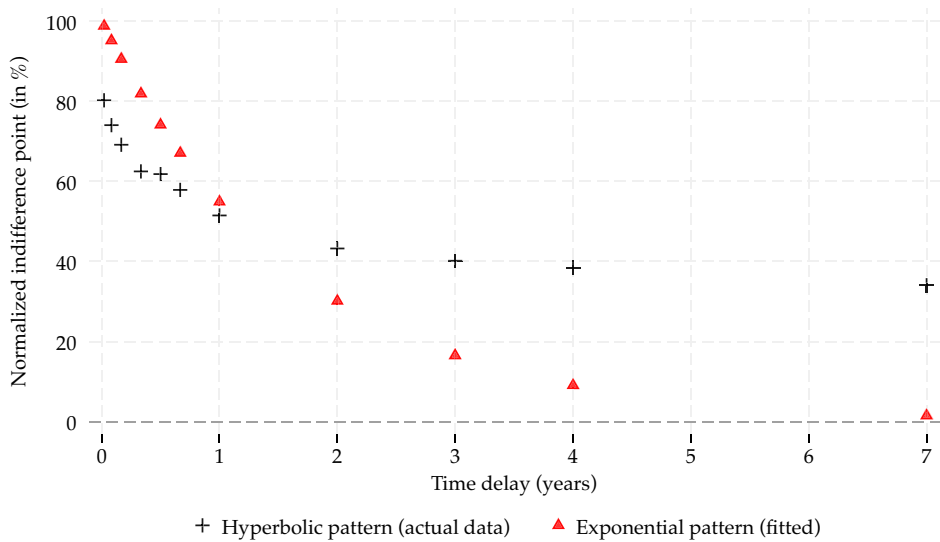


FIGURE 1. Illustration of classical hyperbolic discounting pattern using our experimental data. The pluses show the average earlier payment that makes a decision maker indifferent to a fixed later payment, expressed as a percentage of the later amount (treatment *Delay-M*, see Section 2). The triangles are predicted values from fitting an exponential discounting model to the data.

axis and immediately paid monetary amounts that make people indifferent to these delayed payments on the vertical axis. As the figure shows, decision makers' actual choices tend to be *insensitive* to variations in time delay relative to the pattern predicted by an exponential model fitted to the data. This produces both (i) *extreme impatience* for small delays and (ii) *decreasing per-period impatience* as the payment delays extend further into the future. As a result, structural estimates from the resulting data suggest *apparent present bias*, that is, special attachment to immediate rewards. These and other anomalies from the literature thus boil down to the observation that empirically observed discounting exhibits a hyperbolic pattern rather than an exponential one.¹

Why do these deviations from standard exponential predictions occur, and how should we interpret them? While it is widely understood that decisions over money do not directly reveal underlying time preferences, answering this question is nonetheless of first-order importance because many of the most important intertemporal choices in modern economies are made over money, and many of our standard predictions about these financial choices are made assuming exponential discounting. Moreover,

1. Throughout, we use the term “hyperbolic pattern” as a shorthand for the *empirical* pattern of high short-run impatience coupled with decreasing impatience as delays increase, not as a claim about the specific functional form that best describes that pattern. Likewise, although the word “hyperbolic” is often applied to models of preferences, we do not take a stance on whether money experiments measure true time preferences. Instead we are interested in understanding the reasons for hyperbolic discounting *behavior* over financial flows, which may or may not be driven by non-exponential time preferences.

measures of how people discount future financial flows are highly predictive of a number of real-world behaviors and outcomes, including wealth inequality, income, savings, educational investment, exercise, misconduct in school, tax filing, and food choice (e.g., Ashraf, Karlan, and Yin 2006; Brownback, Imas, and Kuhn 2023; Chabris et al. 2008; Epper et al. 2020; Falk et al. 2018; Martinez, Meier, and Sprenger 2023; Meier and Sprenger 2010; Sunde et al. 2022; Sutter et al. 2013, see overview in [Online Appendix Table A.1](#)). Understanding why these highly predictive measures are typically non-exponential is therefore important for understanding the basis of real-world intertemporal choice.

Two families of explanation. Because these anomalies were first identified in exercises designed to measure people's time preferences, most explanations for them over the last 40 years have involved hypotheses about people's *objective functions*, that is, about their motivational responses to temporal delay. Researchers pursuing this line of explanation have explored the idea that these anomalies occur because people (i) do not in fact have exponential time preferences (e.g., Laibson 1997; Loewenstein and Prelec 1992; O'Donoghue and Rabin 1999), (ii) suffer from temptation effects (Gul and Pesendorfer 2001), (iii) are governed by "multiple selves" with divergent preferences at different dates (Fudenberg and Levine 2006), or (iv) respond in non-exponential ways to transaction costs or the inherent riskiness of the future (e.g., Andreoni and Sprenger 2012; Halevy 2008). All of these accounts rest on the principle that people have special *motivations or objectives* regarding time delays that produce hyperbolic effects.

More recently, a small theoretical literature has considered an alternative possibility: that people fail to exponentially discount because (i) proper discounting requires cognitively expensive information processing and (ii) people respond to the resulting costs and difficulties by substituting imperfect decision processes instead. Importantly, this literature has shown that a number of imperfect evaluation rules of this sort are capable of producing decisions that are *insufficiently sensitive* to time delays, producing hyperbolicity for reasons that have little to do with intertemporal motivations (e.g., Ebert and Prelec 2007; Gabaix and Laibson 2022; Gershman and Bhui 2020; Vieider 2021b). In contrast to *motivational* explanations, we refer to this second class of explanations as *complexity* based explanations, since they are rooted in the idea that the information processing costs of properly aggregating the components of a time discounting problem into a value (i.e., delays and payments) are ultimately responsible for the pattern. For instance, the information processing costs include the difficulty of accessing one's true preferences.

Empirical implications of complexity explanations. The aim of our paper is to identify which of these two broad classes of mechanisms is responsible for the hyperbolic pattern: motivational mechanisms rooted in temporal delay versus cognitive mechanisms rooted in complexity. To do this, we test a series of distinctive implications of complexity-based explanations. To the degree hyperbolicity is driven by complexity, we would expect (#1) hyperbolicity to occur also in computational problems that require similarly complex computations but do not involve actual time delay, (#2) hyperbolicity to be correlated with auxiliary evidence that subjects are using imperfect

heuristics rather than revealing stable preferences (e.g., internal inconsistencies in subjects' choices, or subjects' doubts about the optimality of their own choices), and (#3) hyperbolicity to increase in intensity when intertemporal tasks are made more computationally difficult.

Importantly, these regularities should not similarly occur if hyperbolicity is driven by motivational mechanisms such as time preferences, self-control problems, etc., making them strong tests of the hypothesis that hyperbolicity represents a heuristic response to complexity.

Experimental evidence. We report three series of experimental treatments, designed to test for each of these three implications of complexity-based explanations. We do not view any one of these three approaches as our primary source of evidence, but instead view the three as highly complementary, producing a stronger test together than any one approach would on its own.

To test implication #1, following the design idea of Oprea (2024), we pair standard intertemporal choice problems with what we call “atemporal mirrors” of these same choice problems. In an atemporal mirror, instead of asking subjects to value, for example, \$50 paid in 12 months, we ask them to value \$50 *shrunk 12 times, each time by 4%*. Valuing mirrors requires similar iterative reasoning as exponential discounting, but involves no time delay and therefore cannot produce hyperbolic “discounting” due to standard intertemporal motivations discussed in the literature. Indeed, valuing a mirror in any way other than exponentially is an objective, money-losing mistake.

Nonetheless, we find that the full hyperbolic pattern arises to a similar degree in mirrors as it does in true intertemporal choice. Subjects undervalue payments that are shrunk only a small number of times, but substantially overvalue payments that require a large number of discounting steps. As a result, even though the atemporal mirrors experimentally *induce* an exponential discount function, subjects show clear evidence of diminishing “impatience”, the main signature of the hyperbolic pattern.

Moreover, because in some of our treatments we measure each subject's behavior in both atemporal mirrors and intertemporal choice, we can show that behavior in the former *predicts* behavior in the latter. We find correlations across the two choice problems of 0.34, which means outright valuation mistakes in atemporal mirrors serve as one of the strongest predictors of intertemporal choice measured in the literature. This evidence suggests that behaviors in the two settings are driven by a common behavioral mechanism, which we suggest is rooted in the complexity of information processing, the one property the two decision tasks share.

To test implication #2, we study the relationship between (i) standard money-earlier-or-later decisions and (ii) auxiliary evidence that subjects are responding imperfectly to the information-processing required by the decision task. First, we directly ask subjects whether they believe they optimized by measuring subjects' cognitive uncertainty (CU), the likelihood with which they believe they made a suboptimal decision given their preferences (Enke and Graeber 2023). We find that this measure is highly predictive of hyperbolic discounting in standard intertemporal choice experiments. Indeed, people who are confident that their choices actually reflect their underlying preferences are nearly exponential discounters, meaning the hyperbolic

pattern is concentrated in people who doubt that they actually made optimal choices. Second, we collect rich data on choice inconsistencies, and find that evidence of unstable behavior is similarly predictive of hyperbolicity.

Finally, to test implication #3, we run intertemporal choice experiments in which we describe payoffs and time delays using complicated expressions that require additional information processing. We find that subjects become significantly *more* hyperbolic when we do this, and that auxiliary evidence of complexity responses (e.g., CU) also rises. This again suggests that complexity is a driver of hyperbolicity.

Implications for estimating present bias. It is common in the literature to summarize the hyperbolic pattern of discounting described above using structural estimates of the quasi-hyperbolic $\beta - \delta$ model (Laibson 1997). We show that structural estimates of present bias that are identified from the hyperbolic shape of the empirical discount function alone are severely inflated due to complexity-driven hyperbolicity. For instance, in our atemporal mirrors experiment, we structurally estimate $\hat{\beta} = 0.85$ —an estimate that would conventionally be interpreted as strong evidence of present bias, but here is unambiguously a measure of valuation mistakes. Similarly, in our intertemporal choice experiments over time-dated monetary rewards, we find that structural estimates of present bias are strongly correlated with CU and choice inconsistencies, again suggesting that information-processing constraints spuriously inflate these estimates.

Importantly, however, we find that *causal* estimates of present bias, gathered using front-end delay designs in which both earlier and later payments are moved into the future by a common delay, are not confounded in this way. Thus, our results carry a methodological message: when estimating present bias over money, researchers should use front-end delay designs rather than rely on the hyperbolic shape of the empirical discount function, which confounds complexity-derived error with present bias.

Summary and related literature. Taken together, over a number of distinct empirical approaches—including atemporal mirrors, choice inconsistencies, CU, and experimental complexity manipulations—we document the same story. The classic hyperbolic pattern in intertemporal decisions over money occurs largely because people respond to the difficulty of processing and aggregating delays and payments by using imperfect valuation rules that are *insufficiently sensitive* (“attenuated”) to variation in time intervals. Even if people had exponential discount functions over money, our results suggest that mistakes would produce behavior that looks hyperbolic due to the information-processing required by the valuation problem.

Our paper connects to experimental work that documents various “cognitive effects” in intertemporal choice, such as (i) that time discounting is sensitive to cognitive load, time pressure, and framing (e.g., Dertwinkel-Kalt et al. 2021; Ebert and Prelec 2007; Imas, Kuhn, and Mironova 2022), and (ii) that noise or confusion can spuriously drive estimates of present bias or commitment demand (Carrera et al. 2022; Chakraborty et al. 2017). Also related to us are experimental papers showing that people have difficulty with exponential reasoning, suffering an exponential growth bias (Goda et al. 2015; Stango and Zinman 2009). Our contribution to this body of research is to highlight the implications of complexity for understanding the hyperbolic

TABLE 1. Overview of main experiments.

Sessions	Description	Subjects
<i>Delay & Mirror</i>	18 tasks under <i>Mirror</i> & exactly repeated under <i>Delay</i> (order of treatments randomized)	500
<i>Delay-M</i>	12 delay tasks with elicitations of cognitive uncertainty and choice inconsistencies	645
<i>Voucher-M</i>	12 delay tasks with UberEats vouchers and elicitations of cognitive uncertainty and choice inconsistencies	500
<i>Complex Payments/Delays</i>	12 delay tasks with payoffs or delays described as algebraic expressions	302

pattern in the most widely used experimental paradigm (i.e., money-early-versus-later designs).

More broadly, our paper is linked to an experimental literature on complexity and the non-standard behaviors it induces in other decision domains (e.g., Augenblick, Lazarus, and Thaler 2021; Ba, Bohren, and Imas 2022; Nielsen and Rehbeck 2022; Oprea 2020). The link between complexity and insensitivity that we show drives hyperbolicity, is reminiscent of a seemingly unrelated literature on choice under risk, where an emerging body of work has found that the generic cognitive difficulty of lottery valuation tasks generates an insensitivity to probabilities (e.g., Enke and Graeber 2023; Enke and Shubatt 2023; Frydman and Jin 2023; Khaw, Li, and Woodford 2021, 2022; Oprea 2024; Vieider 2021a), producing systematic patterns like probability weighting. The common thread that runs through these lines of work is that information-processing constraints (and the cognitive noise they produce) cause insensitivity that, in turn, generates famous behavioral anomalies.

Our paper is organized as follows: Section 2 presents our experimental design. Sections 3 and 4 present the results. Section 5 discusses the implications of our results and concludes.

2. Experimental Design

2.1. Basic Setup

Table 1 provides an overview of the components of our experimental design. Following standard methods used in the literature, the core tasks in our experiments are *multiple price lists* that ask subjects to evaluate a payment of x_2 at a time t_2 in terms of dollars paid at an earlier date $t_1 < t_2$. An example of the subject’s decision screen is shown in Online Appendix Figure A.1. In each list, Option A is kept identical in every row, paying x_2 at date t_2 . By contrast, Option B pays an amount x_1 that declines monotonically by \$2 in each row (ranging between x_2 and \$2), at date t_1 . Non-negative discounting entails that subjects choose A in early rows of the list (or, with extreme

preferences, never) and switch to B in some later row (we allow subjects to switch only once in each list).

Under the standard model in economics—the exponential discounted utility model—decision makers in this task discount delayed payments at a constant rate, described by an annual discount factor $\delta = 1 - \gamma$, where γ is approximately the constant discount rate:

$$\gamma = 1 - \left(\frac{x_1}{x_2} \right)^{12/\Delta t} = 1 - e^{-12 \cdot RRR/\Delta t}, \quad (1)$$

where $\Delta t \equiv t_2 - t_1$ is the time delay (all time units are in months), and $RRR/\Delta t = \ln(x_2/x_1)/\Delta t$ is the “interval-adjusted required rate of return” that the decision maker reveals through her choices.² The key prediction of the exponential model is that RRR , and therefore γ will be *constant* as the later payment pushes further into the future. The switching point between earlier and later payment in the experiment yields a direct measure of the RRR and thereby implied annual impatience, γ .

We refer to these choice problems as the *Delay* treatment. In most cases, we randomize (at the subject-list level) the delayed payment $x_2 \in \{\$40, \$42, \dots, \$52\}$. The experimental design includes three main types of price lists. First, “Now Lists”, in which $t_1 = 0$ and t_2 varies across 1/4, 1, 2, 12, 24, 36, 48, and 84 months. Second, “Later Lists”, which are identical to Now Lists except that the earlier payment is slightly delayed: $t_1 = 1$ or $t_1 = 1/4$ months. Finally, “Subadditivity/Front-End Delay (SA/FED) Lists”, in which for some horizon T , we assign subjects lists $(t_1=0, t_2 = T/2)$, $(t_1=T/2, t_2 = T)$ and $(t_1 = 0, t_2 = T)$, maintaining a consistent x_2 across the three lists. We randomly assign T across subjects to be either 8 or 12. Dates t_1, t_2 represent months. Lists from each of these categories are included in every treatment for every subject and randomly ordered at the subject level. Now and Later lists are used primarily to study the anomalies of extreme short-run discounting, decreasing impatience/hyperbolicity, and sub-unitary β . SA/FED Lists are used to measure subadditivity and front-end delay effects, which we discuss in detail at the end of Sections 3 and 4, respectively.

2.2. Tests of Three Implications of Complexity Explanations

The heart of the experiment is a series of treatments that test three main empirical implications of a complexity-based (or information-processing-based) account of the hyperbolic pattern that are not shared by alternative explanations (i.e., time preferences, self-control problems). We discuss each of these three implications below, and use them to propose three different tests of the hypothesis that complexity is responsible for the hyperbolic pattern discussed in Section 1. We view each of these three treatments as equally central to our paper since they rely on three essentially

2. Equation (1) can be derived via the indifference condition $(1 - \gamma)^{t_1/12}x_1 = (1 - \gamma)^{t_2/12}x_2$.

orthogonal empirical approaches with different, mutually reinforcing strengths and weaknesses.

2.2.1. Implication #1: Temporal Invariance. If hyperbolicity is driven by aggregation errors, it should continue to arise even after competing “motivational” explanations have been removed from the choice problem. Unlike motivational explanations like time preferences or self-control problems, information processing-based explanations do not rely in any special way on the actual elapse of time. They instead rely on the fact that intertemporal choice requires decision makers to aggregate multiple pieces of information, which requires intensive information processing. Because they do not depend on time, these sorts of explanations should produce hyperbolicity in decision problems that involve no actual temporal delay, but that require a similar type of reasoning.

Building on this observation, we can construct an immediately paid “atemporal mirror” M_D of choice problem D that replaces payment dates with a sequence of “steps” of payoff discounting. In each step of discounting, the payoff from the previous step is multiplied by an exogenously provided and known fixed factor $\delta < 1$. Thus, an atemporal mirror of D pays a deterministic amount $\delta^t x_1$ or $\delta^t x_2$ immediately. Instead of, for example, valuing a payoff “\$50 in 2 months”, a decision maker evaluating a mirror is asked to value a payoff “\$50 shrunk by δ two times”. An atemporal mirror is therefore nothing more or less than an immediate dollar payment that has been deliberately described in such a way as to require a similar kind of information processing as is required in intertemporal choice.

Because atemporal mirrors involve no actual time delays, hyperbolicity in their evaluation must be driven by mistakes in aggregating the problem components. For instance, potential anomalies cannot be driven by non-exponential time preferences: an atemporal mirror *induces* exponential preferences, making any non-exponential behavior a mistake. Thus, if hyperbolicity is present in such settings, it means that we should expect people to exhibit hyperbolic behavior even if they have exponential preferences.

Experimental implementation. To test this implication, we pair *Delay* tasks with companion problems in which we pay subjects an iteratively discounted version of the stated payoff immediately. In these tasks, discounting occurs through a known, exogenous discount factor, transforming A and B into “atemporal mirrors” of standard intertemporal choice tasks. This is framed to subjects as “shrinking” a payment t times. Each time a payment is “shrunk,” it falls to $\delta < 1$ of its previous value, but a subject must reason through the consequence of this discounting in order to properly value it. The fact that atemporal mirrors are paid immediately is repeatedly emphasized to subjects in the instructions.

A choice list from treatment *Mirror* is displayed in [Online Appendix Figure A.1](#). Each list asks subjects an exactly analogous sequence of binary choice questions as in the corresponding list from the *Delay* treatment. Option A (kept identical in each row of the list) is a dollar payment, paid out immediately but iteratively discounted some number of times. Option B is a dollar payment that involves strictly fewer iterations,

and often none, which mimics an immediate or earlier payment. For example, in one row of a list, subjects are asked to choose between “Option A: \$42 shrunk 12 times” and “Option B: \$2”. We again elicit a standard switching interval to calculate the implied “annual impatience”.

The mirrors we implement include a single step of discounting for each month of discounting in the *Delay* problem it mirrors. Throughout the experiment, we set the per-period $\delta = 0.96$. This particular value was chosen because it corresponds to the estimated monthly discount factor δ in our true intertemporal choice experiments. Our choice to induce a monthly (rather than yearly or daily) discount factor was largely guided by practicality and common sense. First, inducing a yearly discount factor would likely have contributed to participant confusion for delays of less than 1 year. Second, daily discount factors would have led to a very large number of discounting steps for delays of several years.

Every subject from this arm of the study participated in both *Delay* and *Mirror* in a random order. The upside of this within-subjects design is that it allows us to correlate behavior in the two types of problems across subjects. When we are not interested in correlating behavior across treatments, we take care to rule out contamination effects by only analyzing decisions from the treatment that a subject encountered first (the results are very similar when we also include the data from the second-assigned treatment, see [Online Appendix Table A.3](#)).

Because the treatments were designed to be compared to one another, we took great pains to use an identical interface and identical numbers. However, we were also careful to strongly differentiate the payoffs underlying the two treatments from one another using clear instructions. Importantly, to minimize cross-treatment contagion, subjects first assigned to *Mirror* did not know they would later be making intertemporal choices, and vice versa.

The *Mirror* treatment is incentivized using real payments, but the *Delay* treatment is a purely hypothetical elicitation. While the hypothetical nature of the payouts has obvious potential disadvantages, it also confers various advantages, particularly in conjunction with our financially incentivized experiments (see below). First, our use of hypotheticals allowed us to explicitly instruct participants to make their choices assuming that there is no payment risk (a potential confound in incentivized intertemporal choice tasks). Second, hypothetical payments allow us to use some very long time delays (up to “in 7 years”) that are not feasible with real payments, effort, or food vouchers—a constraint that has forced many past researchers to study only shorter delays. On the other hand, evidence suggests that the disadvantages of using hypotheticals may be limited: Reviewing the literature, Cohen et al. (2020) conclude “there is little evidence of systematic differences between RRR in incentivized and unincentivized experiments.”

2.2.2. Implication #2: Correlation With Indices of Bounded Rationality. If complexity drives hyperbolicity, then hyperbolicity should be correlated with independent evidence that subjects are not using optimal choice rules and are making aggregation mistakes. Thus, another approach to evaluating information-processing

explanations is to directly measure behavioral signatures of such errors in standard intertemporal choice problems, and to study whether this evidence predicts the presence and severity of hyperbolic discounting.

The literature has proposed two empirical indicators that a decision was made using an imperfect (i.e., heuristic or noisy error-prone) choice rule: self-reported CU and choice inconsistencies. Our empirical strategy is to study to what degree these behavioral signatures of error-prone decision-making predict the incidence and severity of anomalies.

Experimental implementation. To test this implication, we run another pair of treatments, which are variations on the standard choice setting described above in which we gather auxiliary evidence that subjects are using decision rules that are error-prone. In both of these treatments, we measure the following objects.

- **Cognitive Uncertainty.** Adapting the methodology from Enke and Graeber (2023), after each choice list, we measure CU as the subject's subjective probabilistic belief that their true valuation of the later payment is contained in their stated switching interval:

Your choices on the previous screen indicate that you value $\$x_2$ in t_2 somewhere between $\$a$ and $\$b$ in t_1 . How certain are you that you actually value $\$x_2$ in t_2 somewhere between $\$a$ and $\$b$ in t_1 ?

Participants answer this question by selecting a radio button between 0% and 100%, in steps of 5%; see [Online Appendix Figure A.2](#). We interpret this question as measuring the participant's awareness that their decision procedure is noisy or heuristic.³ The measure is not incentivized.

- **Choice Inconsistencies.** A standard way of measuring the noisiness of subjects' decision procedures is choice inconsistency in repetitions of the same choice problem (randomly interspersed throughout the experiment). In our study, each subject completes two randomly selected choice lists twice. We use this to generate a binary indicator that equals one if the subject's decisions on the two repeated trials are different from each other (we also verify that our results continue to hold if we instead compute the absolute difference between the two decisions as our measure of inconsistency).

We collected these two pieces of data in two different treatments. In *Delay-M*, we again use hypothetical monetary payments, which allows us to study multi-year delays. The *Voucher-M* treatment is identical in most respects, except (i) that we actually pay subjects for their choices using UberEats food delivery vouchers and (ii) we do not study delays of more than 1 year (for feasibility reasons). UberEats vouchers are usable starting at date t_1 or $t_2 \leq 12$, respectively; these vouchers are valid for a period of

3. We ensure that subjects do not misunderstand the question as referring to *external* uncertainty that they may not actually receive the reward. To this effect, our experiments include a comprehension check question that directly asks participants to indicate whether the CU elicitation question asks about (i) the subject's subjective probability of actually receiving the money or (ii) their certainty about their own valuation, given that they know they will receive the money with certainty.

only 7 days from the starting date, which reduces (but does not eliminate) fungibility concerns. Subjects again complete multiple price lists, except that all payments refer to UberEats vouchers (of value between \$40 and \$50). Participants' vouchers were directly credited to their personal UberEats accounts within 10 hours of completion of the study, such that subjects did not have to actively claim the voucher. The vouchers were always visible in their accounts; they could just not be used before the validity period. Because participants could always view vouchers in their account within a few hours of the study regardless of the precise validity period, there is no differential payment risk across vouchers with different time delays. Participants received automatic reminders 24 hours before a voucher became valid and 24 hours before it expired.

2.2.3. Implication #3: Sensitivity to Complexity Manipulations. If hyperbolicity is a response to complexity, it should increase when the complexity of the choice problem exogenously increases. In particular, it should increase if we deliberately describe payouts and delays in especially complicated ways, requiring more intensive information processing on the part of subjects to aggregate them into a decision. To the extent the hyperbolic pattern is a consequence of costly or difficult information processing, we would expect the pattern to intensify in response to overt efforts to increase the required information processing.

Experimental implementation. In treatment *Complex Payments/Delays*, we replicate treatment *Delay-M*, but for one subset of subjects ($N = 153$), we express all of the payoffs in the price list as an algebraic expression (e.g., \$40 is described as “ $(4 \cdot 8/2) + (8 \cdot 9/2) - 12$ ”). For another subset ($N = 149$), we express all dates in the price list as algebraic expressions (e.g., 1 year is described as “in $(6 \cdot 2/3 - 3)$ years AND $(3 \cdot 6/2 - 9)$ months AND $(5 \cdot 4/2 - 10)$ days”). These interventions are always paired with time constraints of 25 seconds to make the relevant information processing constraints more likely to bind. In these treatments, as in treatment *Delay-M*, we collect both choice inconsistency and CU measures, allowing us to also verify that these measures are causally influenced by and responsive to complexity.

2.3. Procedures

All experiments were conducted on Prolific. [Online Appendix E](#) contains details on experimental instructions, visual display, and screening questions used.⁴

Subjects in the *Mirror & Delay* sessions were paid a \$6 base payment and had a 20% chance of being paid a bonus based on their choice from a randomly selected list and row of Mirror (or from a separate risk elicitation we included in our sessions). In *Delay-M*, subjects earned a flat \$4.50 payment. In *Voucher-M*, subjects received a \$4

4. All experiments except *Delay & Mirror* were pre-registered and originally reported in a different working paper: see all details in Enke & Graeber (2021). In that working paper we also report on an experimental manipulation of cognitive load, which produces very similar results as treatment *Complex Payments / Delays*.

base payment and voucher payments from a randomly selected list and row with a 25% chance.

3. Complexity and Hyperbolic Discounting

Testing implication #1: evidence from atemporal mirrors. We begin by examining whether hyperbolic discounting appears in *Mirror* and comparing its magnitude to hyperbolicity in *Delay*. In analyzing this data, it is important to emphasize that there are at best weak reasons to expect similar “patience” levels in the two treatments. In *Mirror*, subjects face an induced discount factor of 0.96; in *Delay*, choices depend on subjects’ individual discount factors, which may differ at the individual level from 0.96. Our focus will therefore be on comparing the severity of *hyperbolicity* rather than comparing measured patience levels.

The top panels of Figure 2 provide an overview of the raw data by plotting, for both the *Delay* and *Mirror* treatments, the average switching point (expressed as a percentage of the “later” payment, x_2) as a function of the time interval or the number of discounting iterations.⁵ Recall that in an exponential discounting framework with linear utility, these normalized switching points correspond to $x_1/x_2 = \delta^{\Delta t}$. For the *Mirror* treatment, we overlay the indifference point that a payoff-maximizing subject would choose given the induced “monthly discount factor” ($\delta = 0.96$). Both panels pool data from Now Lists (the earlier date is immediate or paid with no discounting in *Mirror*) and from Later Lists (the earlier date is in 1 month or after one step of discounting) because, as Online Appendix Figure A.3 shows, the results look very similar for both types of lists.

The bottom panels of Figure 2 transform the data in a straightforward way by computing implied annual impatience, $\hat{\gamma} = 1 - (x_1/x_2)^{12/\Delta t} = 1 - e^{-RRR \cdot 12/\Delta t}$, see equation (1).

The figures show that, as expected, we replicate the hyperbolic pattern in our control *Delay* treatment. First, our subjects show very high impatience over short horizons. Second, indifference payments are a highly compressed function of the time interval. The bottom panels show that this insensitivity implies that implied annual impatience is sharply decreasing in the length of the interval.

Our main finding is that both of these patterns also appear as clear mistakes in the *Mirror* treatment. First, as in *Delay*, subjects in *Mirror* show extreme discounting over the first few steps of discounting, even though there is no actual delay in these problems and even though these subjects are incentivized to maximize an exponential discount function. Second, as Figure 2 shows, subjects’ indifference payments in the atemporal mirrors are insufficiently sensitive to the number of discounting steps relative to the experimentally induced discount factor. As in *Delay*, this insensitivity implies a strong decrease in implied per-period impatience as the number of discounting steps

5. We approximate switching points by computing the midpoint of the switching interval.

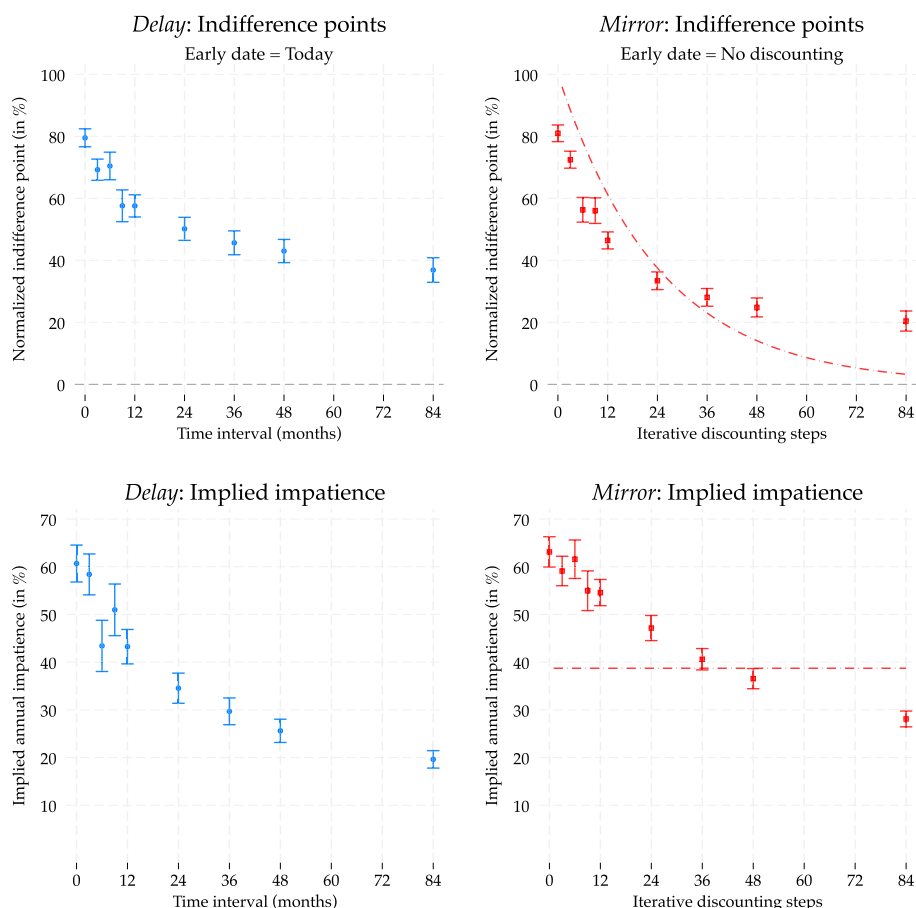


FIGURE 2. Top panels: Average normalized indifference points by time interval (*Delay*) or number of iterations (*Mirror*). Bottom panels: Average implied annual impatience $\hat{\gamma}$ by time interval or number of iterations. Left panels show *Delay* treatment (4,572 decisions from 254 participants). Right panels show *Mirror* treatment (4,428 decisions from 246 participants). In the *Mirror* panels, the dashed line represents payoff-maximizing decisions. The time interval in months and the number of iterations are rounded to the nearest multiple of three. Whiskers show 95% confidence intervals, computed based on clustering at the subject level.

increases. However, as the figure highlights, in *Mirror* this is a clear, money-losing mistake: Subjects' average implied "impatience" in Figure 2 is located above the normative benchmark for a few iterations but below it for many iterations.

As is clear from the figures, not only does the full hyperbolic pattern occur in *Mirrors*, but it also appears with similar severity as in *Delay*. To verify this formally, [Online Appendix Table A.2](#) presents regression evidence. In *Delay*, for each additional year, implied annual impatience decreases by 5.6 percentage points (pp). In *Mirror*, that effect is 4.8 pp, meaning that decreasing impatience in *Mirror* is 86% as strong as in *Delay*.

RESULT 1. *Subjects exhibit extreme short-run impatience and decreasing impatience when evaluating atemporal mirrors just as they do when evaluating delays. For mirrors, these are clear misvaluations.*

Testing implication #2: correlations with measures of bounded rationality. Next, we examine the relationship between hyperbolicity and CU and choice inconsistencies using data from treatments *Delay-M* and *Voucher-M*. We show the results using figures in this section and verify that all findings are statistically significant using econometric analyses in [Online Appendix Table A.4](#).

The data from the *Delay-M* and *Voucher-M* treatments strongly suggest that subjects' tend to use imperfect valuation rules to make their choices. In *Delay-M*, 75% of all decisions are associated with strictly positive CU, and 60% of all repeated decisions show strictly positive inconsistency. In *Voucher-M*, the corresponding frequencies are 83% and 60%. Thus, most subjects express doubt about the optimality of their own choices, and most make unstable choices, both of which suggest that most subjects are not using optimal choice rules to value delayed payments.

To study whether these indices of imperfect choice predict hyperbolicity, we break the data down according to the presence/absence of CU and choice inconsistency. The left-hand panels of Figure 3 show raw data for the *Delay-M* treatment by plotting the relationship between normalized indifference points (in percent) and time intervals. In each case, we split results based on the presence or absence of (i) measured CU in the decision (top panel) or (ii) choice inconsistency in the decision (bottom panel). The corresponding right-hand panels transform these data (as in the previous section) by computing implied annual impatience $\hat{\gamma}$. All panels pool the data for Now and Later lists (the results are very similar looking at each of them separately). Figure 4 shows analogous results for the incentivized *UberEats* voucher experiments.

The top panels show that decisions accompanied by strictly positive CU are considerably *less sensitive* to variation in the time delay, making them look considerably more hyperbolic. This has direct implications for each component of the hyperbolic pattern. First, CU is strongly predictive of extreme short-run impatience. Second, implied annual impatience decreases much more rapidly in the time interval for cognitively uncertain than for cognitively certain subjects. For instance, going from $\Delta t \approx 1$ to $\Delta t \approx 84$ months, the implied annual impatience drops by a factor of 4.5 when $CU > 0$, but only by a factor of 2 when $CU = 0$. In treatment *Delay-M*, this means that cognitively uncertain participants act as if they are *less* patient over relatively short horizons, yet *more* patient over relatively long horizons.⁶

Strikingly, the strong link between CU and insensitivity to delays is also present in within-subject comparisons. To show this, we normalize the CU data to have mean zero and standard deviation one for each subject, and then look at whether this pure

6. These results do not hinge on splitting the sample into decisions with zero versus strictly positive CU. To show this, we split the sample into CU quartiles. We find that the effect of the time interval on decisions continuously decreases (in absolute terms) as CU increases; see [Online Appendix Figure A.4](#).

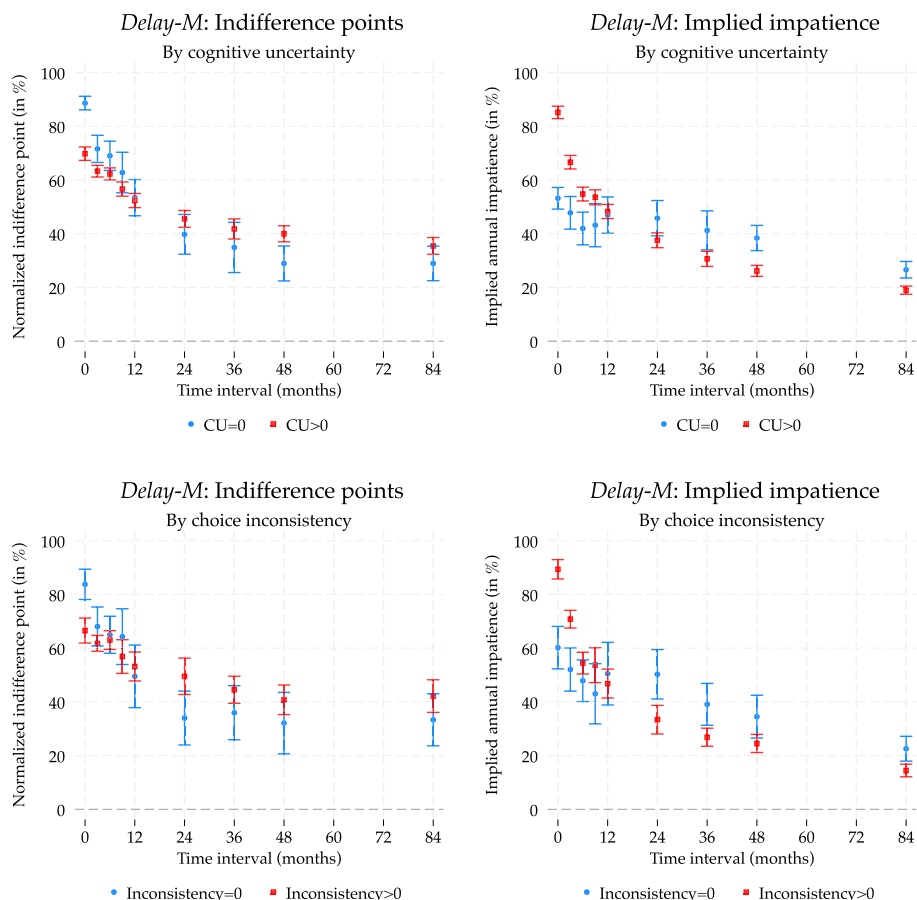


FIGURE 3. Normalized indifference points (left panels) and implied annual impatience (right panels) as a function of the time interval. The top panels include all decisions from *Delay-M*, and we split the sample according to whether or not a choice is associated with strictly positive CU (7,740 decisions by 645 subjects). The bottom panels include data from all decisions in *Delay-M* that were elicited twice (two repeated problems per subject for a total of 2,580 decisions from 645 subjects), and we split the sample according to whether or not decisions differed in a set of repeated choices. Time intervals are rounded to nearest multiple of 3 months. Whiskers show 95% confidence intervals, computed based on clustering at the subject level.

within-subject measure still predicts choices. [Online Appendix Table A.5](#) provides an affirmative answer.

The bottom panels of Figures 3 and 4 show that we find directly analogous results if we replace CU with choice inconsistency as our measure of the use of an imperfect choice rule.

What fraction of decreasing impatience is driven by valuation mistakes, according to this analysis? To quantify this, we compare the magnitudes in two sub-samples: (i) decisions that are associated with no CU and no choice inconsistency versus (ii)

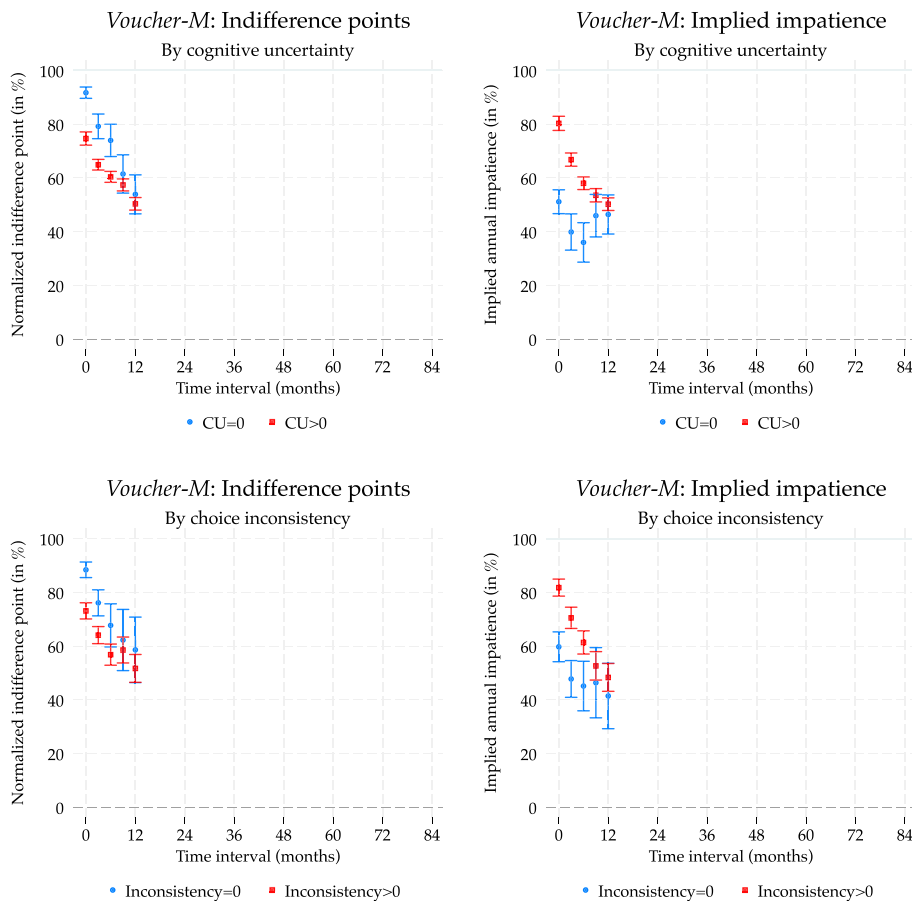


FIGURE 4. Normalized indifference points (left panels) and implied annual impatience (right panels) as a function of the time interval in *Voucher-M*. The top panels include all decisions, and we split the sample according to whether or not subjects indicate strictly positive CU (6,000 decisions from 500 subjects). The bottom panels include data from all decisions that were elicited twice (two repeated problems per subject, for a total of 2,000 decisions from 500 subjects), and we split the sample according to whether or not decisions differed in a set of repeated choices. Time intervals are rounded to the nearest multiple of 3 months. Whiskers show 95% confidence intervals, computed based on clustering at the subject level.

decisions that reflect either strictly positive CU or choice inconsistency. We examine how strongly implied annual impatience increases in the evaluated time interval, akin to the regressions in Online Appendix Table A.4. We find that in the sample with no CU and no choice inconsistencies, the magnitude of decreasing impatience is only 10% of that in the comparison sample. This suggests that at least 90% of decreasing impatience is driven by valuation errors. This is strikingly similar, quantitatively, to the decomposition computed by comparing decreasing impatience in atemporal mirrors and time intervals in the preceding section.

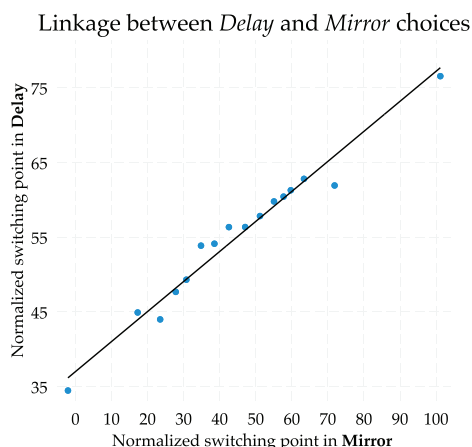


FIGURE 5. Binned scatter plot of normalized indifference points in structurally identical choice problems in *Delay* and *Mirror*. Partial correlation plot, controlling for fixed effects for each choice list type (each possible combination of t_1 and t_2). Based on 18,000 decisions by 500 subjects. The partial correlation is $r = 0.34$.

RESULT 2. *Short-run impatience and decreasing impatience are strongly predicted by auxiliary evidence of valuation errors.*

Linkage between atemporal and intertemporal decisions. Next, we provide evidence that valuations in atemporal mirrors and true intertemporal choice are linked, suggesting a shared mechanism of valuation errors. To establish this, we show that anomalies in *Mirror* and *Delay* are linked at the individual level. To do this, we leverage the within-subjects design of treatments *Delay* and *Mirror* to examine the within-subject relationship between behaviors across the two treatments. If there is a common behavioral mechanism behind the anomalies across treatments, behavior in the two cases should be correlated with each other.

We link subjects' decisions in those choice problems that are direct mirror images of each other, such as "\$40 in 6 months" versus "\$40 shrunk 6 times". Thus, we compute a correlation coefficient for (500 subjects $\times$ 18 unique problems $\times$ 2 treatments $\Rightarrow$) 18,000 observations. In doing this, we take care to net out that component of the correlation that is mechanically driven by the fact that for longer intervals or a higher number of iterations, subjects should be expected to state lower valuations. Thus, we compute the partial correlation between decisions, netting out fixed effects for each unique problem type (each possible combination of t_1 and t_2). The resulting correlation captures how similar subjects' behavior is across the two treatments, holding fixed the nature of the choice problem.

We find a partial correlation of $r = 0.34$ ($p < 0.01$) and illustrate this relationship in Figure 5. This correlation is remarkably high given that the absence of time preference-based variation in the *Mirror* treatment should produce correlations *close to zero* for rational decision makers. Instead, behavior in *Mirror* produces one of the

strongest predictors of intertemporal choice documented in the literature (Cohen et al. 2020).⁷

This result suggests that true intertemporal choice is partly driven by a behavior mechanism shared with valuations of atemporal mirrors. Since variation in behavior in mirrors can only be driven by mistakes, this common mechanism must be valuation errors. Thus, these results suggest that valuation errors are a major driver of true discounting behavior.

RESULT 3. *Across subjects, valuations in intertemporal choice problems are correlated with the valuation of atemporal mirrors, suggesting that behavior across the two domains is driven by a common mechanism (valuation errors).*

This conclusion is reinforced by the fact that choice inconsistencies predict hyperbolicity in atemporal mirrors in a manner that is strongly parallel to true intertemporal delays. In particular, in our *Mirror* and *Delay* treatments, discussed above, we repeated one randomly selected choice list for each subject. As we show in [Online Appendix Table A.6](#), we find that choice inconsistencies are strongly predictive of “short-run impatience” and “decreasing impatience” in mirror valuations, just as they are in true intertemporal decisions. Thus, there is a strong connection between our evidence for implications #1 and #2.

Testing implication #3: manipulation of task complexity. In treatment *Complex Payments/Delays* ($N = 153$), we express all of the payoffs in the price list as an algebraic expression (e.g., \$40 is described as “ $(4 \times 8/2) + (8 \times 9/2) - 12$ ”). For another subset ($N = 149$), we express all dates in the price list as algebraic expressions (e.g., 1 year is described as “in $(6 \times 2/3 - 3)$ years AND $(3 \times 6/2 - 9)$ months AND $(5 \times 4/2 - 10)$ days”). These interventions are always paired with time constraints of 25 seconds to make the relevant information processing constraints more likely to bind.

We report two key findings. First, we show that this intervention significantly increases both of our proxy measures of boundedly rational choice. Average CU rises from 21.7% in *Delay-M* to 35.2% for *Complex Payments/Delays*; choice inconsistency rises from 60.4% in *Delay-M* to 67.2% in *Complex Payments/Delays* (both comparisons are statistically significant at least at the 5% level, see [Online Appendix Table A.7](#)). This establishes a strong connection between complexity implication #2 and #3.

Next, we show that this manipulation simultaneously intensifies intertemporal choice anomalies. As [Figure 6](#) shows, the decisions of subjects in *Complex Payments/Delays* evince stronger short-run impatience and diminishing impatience than those of subjects in *Delay-M* (see [Online Appendix Tables A.7](#) and [A.8](#) for regression evidence). Thus, the manipulation of task difficulty has the same effects as the patterns we observed correlationally for choice inconsistency and CU.

7. For instance, the correlation between valuations of atemporal mirrors and delays is in the same ballpark as the correlation documented between identical intertemporal list choices made by the same subjects in elicitations delivered several months apart (Meier and Sprenger 2015).

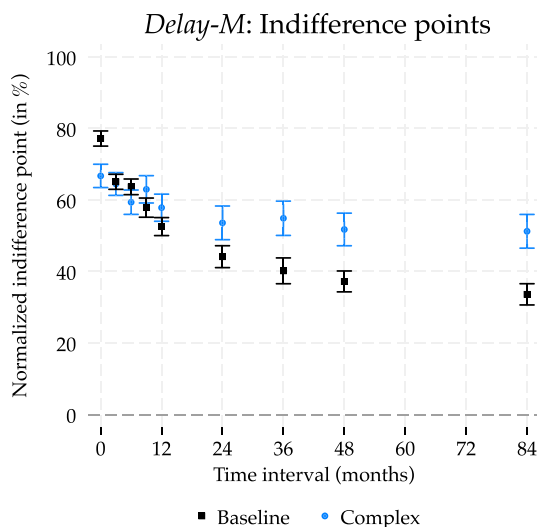


FIGURE 6. Normalized indifference points as a function of the time interval (rounded to the nearest multiple of 3 months) in *Delay-M* and the two *Complex* manipulations, pooled for ease of readability (11,364 decisions from 947 subjects). Whiskers show 95% confidence intervals, computed based on clustering at the subject level.

RESULT 4. *Short-run impatience and decreasing impatience become significantly more pronounced when complexity is exogenously increased.*

Mechanism: complexity and insensitivity. The key takeaway from the preceding analysis is that information-processing constraints cause hyperbolicity by inducing an insensitivity of decisions with respect to the delay. In [Online Appendix D](#), we provide further evidence for this mechanism by studying the role complexity plays in *subadditivity effects*. Subadditivity experiments show that impatience over a single time interval (t_1, t_3) tends to be considerably smaller than the total impatience people reveal when they are asked to make two decisions, one over interval (t_1, t_2) and one over (t_2, t_3), with $t_1 < t_2 < t_3$ (Read 2001). These transitivity violations are direct evidence of insensitivity (i) because they involve people treating shorter intervals too much like they treat a longer interval, and (ii) because this cannot be confounded with non-stationarities in discounting since the decisions involve the comparison of nested intervals. In [Online Appendix D](#), we show that subadditivity effects are strong in our data and that they appear to be *entirely* driven by complexity. In particular, we show that subadditivity only occurs for cognitively uncertain subjects, increases dramatically with our complexity manipulations, and appears to an identical extent in atemporal mirrors as in true intertemporal choice. Since the insensitivities measured by subadditivity effects have been linked to hyperbolicity by the prior literature (Read 2001), this serves as further evidence that hyperbolicity in the empirical discount

function over money is primarily a consequence of complexity-driven insensitivities to delays.

What makes intertemporal choice problems complex? One possibility, *ex ante*, is that it is difficult to introspectively evaluate or calculate one's own time preferences. However, the fact that hyperbolic discounting arises with near-full strength in atemporal mirrors instead suggests that complexity is also partly driven by the difficulty of aggregating the multiple constituent components of an intertemporal choice problem. [Online Appendix C](#) provides some evidence in support of this idea. There, we document that both CU and the variance of subjects' decisions strongly increase in the interval length or the number of iterations required to discount (i.e., the number of aggregation steps that need to be performed). This evidence tentatively suggests that repeated discounting is cognitively costly, producing noise that increases in the number of iterations required to discount a reward.

4. Complexity and Estimates of Present Bias

Our findings suggest that the hyperbolic shape of the empirical discount function in monetary rewards is largely driven by mistakes in aggregating the components of an intertemporal choice problem. What guidance can this provide for efforts to measure *present bias*, perhaps the most often measured empirical object in the intertemporal choice literature?

Structural estimates of present bias. We begin by measuring present bias using the approach taken by structurally estimating the parameters of a $\beta - \delta$ model. Intuitively, in these model estimations, present bias is identified off the hyperbolicity of the empirical discount function, including especially the excess degree of short-run impatience not captured by the estimated exponential discounting parameter, δ . Because we know from the previous section that this hyperbolicity is largely driven by complexity-driven valuation errors, there are strong reasons to hypothesize that structural estimates of β will, likewise, be confounded by complexity and mistakes. Because in this section we will frequently distinguish between structural and experimental estimates of present bias, we will denote structural estimates by $\hat{\beta}_{ST}$.

We first examine whether there is evidence for $\beta_{ST} < 1$ in our *Mirror* treatment, in which exponential discounting is experimentally induced and true present-biased motivations are therefore removed by design. Recall that in all of our experiments, a subject is asked to state an amount x_1 in t_1 that makes her indifferent to x_2 in t_2 . In a $\beta - \delta$ model with linear utility, we, hence, have

$$\delta^{t_1} \cdot x_1 = \beta_{t_1=0} \cdot \delta^{t_2} \cdot x_2. \quad (2)$$

We can estimate this model at the population level, amended by a mean-zero error term (see [Online Appendix Table A.8](#)). In *Mirror*, we estimate $\hat{\beta}_{ST} = 0.85$ (*s.e.* = 0.01)

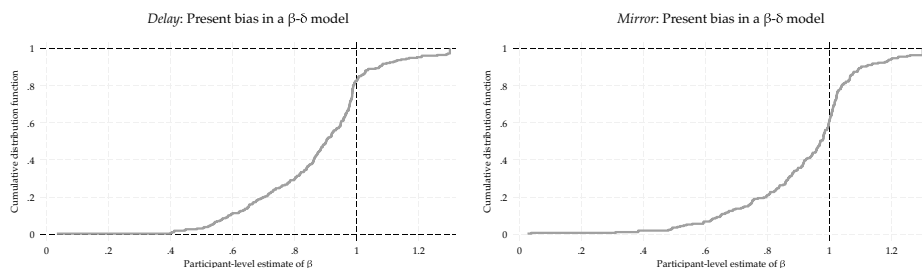


FIGURE 7. Empirical CDFs of individual-level estimates of a $\beta - \delta$ model (equation (2)) in *Delay* ($N = 254$) and *Mirror* ($N = 246$), using first-assigned treatment only. Non-linear least squares estimation based on 18 decisions from each individual. For ease of readability, we top-code estimates of $\hat{\beta}$ greater than 1.3 to 1.3.

and $\hat{\delta} = 0.96$ ($s.e. = 0.01$).⁸ Valuation mistakes alone, therefore, induce behavior that *looks like* present bias under the lens of standard estimation approaches. Intuitively, the reason for this result is that decisions in the *Mirror* treatment have a hyperbolic shape that gets attributed to a sub-unitary β_{ST} . Indeed, our estimates recover the true, induced δ of 0.96, suggesting that most of the distorting effects of valuation errors appear in the spurious estimate of β .

If errors indeed confound structural estimates of present bias, we should also see that—in traditional intertemporal choice experiments—decisions that are associated with stronger CU and choice inconsistencies are associated with more pronounced estimated present bias. To investigate this, we turn to the data from the *Delay-M* treatment. As is well understood in the literature, individual-level heterogeneity in discount factors renders population-level estimates of β_{ST} potentially biased (Weitzman 2001; Jackson and Yariv 2014). Thus, we estimate equation (2) separately for each subject. Figure 7 visualizes the result of the individual-level estimation. Figure 8 shows a binned scatter plot of the $\hat{\beta}_{ST}$ estimates against a summary index of signatures of valuation errors (the first principal component of CU and choice inconsistency). Estimated present bias is strongly concentrated in subjects with evidence of valuation errors (Spearman's $\rho = -0.28$, $p < 0.01$).⁹ Online Appendix Figure A.5 shows that quantitatively almost identical results hold when the estimation

8. Note that because all subjects are induced to have the same time preferences, estimates of β in atemporal mirrors do not run afoul of the aggregation concerns raised in the literature (Jackson and Yariv 2014; Weitzman 2001). Estimates at the individual level corroborate this result. As shown in Figure 7, for the majority (61.4%) of subjects in *Mirror*, we estimate $\hat{\beta}_{ST} < 1$.

9. Population-level estimates deliver similar results on how $\hat{\beta}$ varies with signatures of valuation errors. Online Appendix Table A.8 reports the results. For example, for $CU = 0$, we estimate $\hat{\beta}_{ST} = 0.87$, while for $CU > 0$, we get $\hat{\beta}_{ST} = 0.72$. In contrast, the estimates of δ are always very similar across the different sub-samples, suggesting that (as with our estimates from *Mirror*), valuation errors mostly influence the present bias β term in estimates of these models. These results show that *even if aggregation was not an issue*, valuation mistakes would still bias the estimation of present bias.

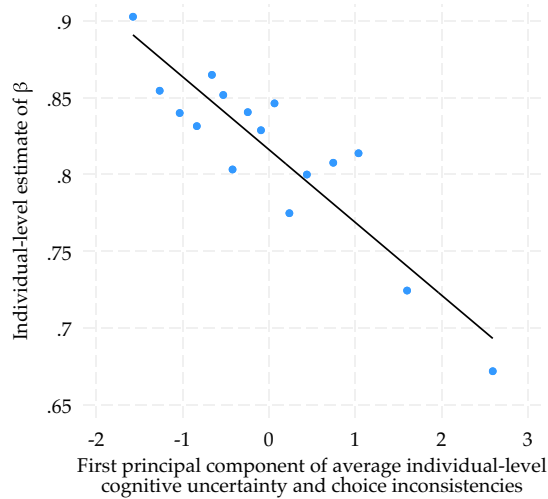


FIGURE 8. Binned scatter plot of individual-level $\hat{\beta}_{ST}$ in equation (2) against first principal component of subject-level average CU and choice inconsistencies. Based on 643 subjects in *Delay-M*. The figure excludes two subjects for whom we estimate $\hat{\beta} > 2$.

accounts for utility curvature (measured through separate lottery choice lists at the end of the experiment). In combination with the result of “present bias” in the atemporal mirrors, this strongly suggests that conventional structural estimates of present bias to a great extent pick up mistakes in valuation.

Note that while empirical estimates of $\beta_{ST} < 1$ are common in the literature using the MEL paradigm (Cohen et al. 2020), recent reviews suggest that this might partly depend on the elicitation method. For example, Imai, Rutter, and Camerer (2021) show that experiments that rely, unlike ours, on the convex time budgets method often find $\beta_{ST} \approx 1$. Although understanding these elicitation effects is beyond the scope of the present study, we emphasize that the apparent dependence of these estimates on the specifics of the elicitation method resonates with our finding that intertemporal anomalies arise in part from complexity-driven heuristics, since heuristics are often “overfit” to details of the choice tasks (see Oprea (2024) for evidence on this point in the domain of risk).

Causal Estimates of Present Bias. A standard way of causally identifying present bias in the literature is by measuring *front-end delay effects* (direct measurements of stationarity violations). In experimental documentations of these effects, subjects reveal lower discounting in evaluating $(t_1 + d, t_2 + d)$ than in (t_1, t_2) , for $d > 0$. Some of our tasks feature such a front-end delay structure (with $t_1 = 0$ and d randomized between 4 and 6 months across subjects).

Table 2 summarizes the evidence on the link between valuation errors and front-end delay effects in our data. Columns (1) and (2) show that we find a statistically significant front-end delay effect in the *Delay* treatment but the *opposite* effect in the *Mirror* treatment. Thus, the mirror data provide no evidence that the front-end delay

TABLE 2. Complexity and front-end delay effects.

Phenomenon: Treatment:	Dependent variable:					
	Implied annual impatience (in %)					
	Front-end delay					
	Delay (1)	Mirror (2)	Delay-M (3)	Delay-M (4)	Voucher-M (5)	Voucher-M (6)
1 if front end delay	-4.24** (1.85)	3.79** (1.69)	-3.07*** (0.99)	-2.51 (1.53)	-4.11*** (1.09)	-7.23*** (2.12)
Cognitive uncertainty				0.38*** (0.06)		0.38*** (0.07)
1 if front end delay × Cognitive uncertainty				-0.058 (0.05)		0.070 (0.07)
Payment amount FE	Yes	Yes	Yes	Yes	Yes	Yes
Task set FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	508	492	2,393	2,393	2,337	2,337
R ²	0.07	0.02	0.02	0.07	0.02	0.08

Notes: OLS estimates, robust standard errors (in parentheses) are clustered at the subject level. The data are restricted to problems that have a front-end delay structure. Task set FE are fixed effects for each pair of tasks that have a front-end delay structure. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

effect is an outgrowth of valuation errors. If anything, our results suggest that such errors might even work against the identification of these effects.

Columns (3)–(6) show the results for treatments *Delay-M* and *Voucher-M*.¹⁰ Again, we find evidence for the presence of front-end delay effects in intertemporal decisions. Importantly for our purposes, however, these effects are entirely uncorrelated with CU, suggesting again that they have little to do with complexity and mistakes. Finally, we show in [Online Appendix Table A.7](#) that the experimental complexity manipulation described in Section 3 does not amplify front-end delay effects, providing a third piece of evidence that front-end delay effects have little to do with the use of error-prone decision rules.

Thus, the atemporal mirrors, the measure of CU, and the experimental complexity manipulation all suggest that non-stationarity (e.g., “true” present bias) is *not* driven by complexity-caused mistakes. Complexity produces insensitivity to the length of the delay, but, since delays are held constant across choice problems in front-end delay designs, they are robust to the effects of complexity we document here. At the same time, in our data, the severity of present bias derived from such designs is much smaller than present bias estimated structurally from the shape of the discount function.

Reconciling structural and experimental estimates. Intuitively, the large differences we find between structurally estimated and front-end-delay-estimated present bias arise because the empirical discount function is substantially more hyperbolic than the magnitude of front-end delay effects would imply. Indeed, this discrepancy between the magnitude of front-end delay effects and of hyperbolic discounting is also highlighted in the review of Cohen et al. (2020). They call the coexistence of strongly decreasing impatience and relatively small front-end delay effects “contradictory patterns”. Our results show that valuation errors are the main driver of this discrepancy because they produce hyperbolicity but not front-end delay effects.¹¹

To sum up, our results suggest that the severely inflated structural estimates of present bias over money reflect model misspecification: Conventional estimates of a $\beta - \delta$ model do not account for valuation mistakes, causing error-induced hyperbolicity of the discount function to be spuriously attributed to β .

RESULT 5. *Structural estimates of present bias over money (that do not rely on causal experimental designs) are severely biased due to model misspecification resulting from ignoring the influence of valuation errors. On the other hand, treatment-based estimates of present bias are unconfounded by complexity.*

10. Recall that we elicited only two randomly selected decisions per subject repeatedly. Given that these repeated decisions do not always occur for the choices in the SA/FED lists, we do not have access to a task-level measure of choice inconsistency that can be used to shed light on subadditivity or front-end delay effects. By contrast, the CU measure is available for each decision a subject makes.

11. Cohen et al. (2020) infer as much, attributing the remainder of hyperbolicity to the insensitivities described by subadditivity effects.

TABLE 3. Summary of results across experiments.

	Short-run impatience	Decreasing impatience	Sub- additivity	Front-end delay effect	Estimated present bias
Present in atemporal mirrors?	✓	✓	✓	–	✓
More pronounced with cognitive uncertainty?	✓	✓	✓	x	✓
More pronounced with choice inconsistency?	✓	✓	n/a	n/a	✓
More pronounced in difficult problems?	✓	✓	✓	–	✓

Notes. “✓” means that an anomaly is present/more pronounced, “x” that it is not present/not more pronounced, and “–” that the opposite is present/the anomaly is less pronounced. “n/a” means that data limitations do not allow us to assess a relationship.

5. Discussion

Table 3 summarizes the results from our paper across all of our treatments. Regardless of how we operationalize and measure complexity effects and the mistakes they produce (through atemporal mirrors, choice inconsistency, CU, and exogenous treatment interventions), we consistently find that mistakes are strongly associated with short-run impatience, decreasing impatience, subadditivity, and structural estimates of present bias. Indeed, across methods, we find strikingly similar *quantitative* evidence that each of these signatures of hyperbolicity are attributable to such errors. We interpret this as evidence that intertemporal tradeoffs over money generate behavioral distortions in large part because they require a difficult cognitive act, which produces an insensitivity of decisions to delays. In contrast, treatment-based estimates of front-end delay effects (“true” present bias) are unconfounded by complexity.

We conclude the paper by discussing broader implications and methodological lessons. When behavioral economists observe choice behavior that is at odds with the neoclassical model, they often (though not always) attribute it to non-standard objective functions and preferences. Yet as this paper and other recent work highlight, surprising choice behavior is often driven by information-processing constraints and the effects these constraints have on economic choices. This has potential implications for (i) the interpretation of intertemporal choice, policy, and welfare; (ii) the experimental methodology we deploy to study human behavior; and (iii) our understanding of the link between anomalies across different decision domains.

Implications for intertemporal choice. Our finding that intertemporal choice anomalies involving money reflect heuristic responses to complexity rather than motivational responses to time delays has several potential implications. First, and most importantly, it calls into question normative interpretations of behaviors like hyperbolic discounting or present bias over money, and therefore is potentially important for the way we conduct behavioral welfare economics. Our results suggest that we should expect standard intertemporal choice anomalies to arise even among decision makers whose normative preferences are exponential. This, in turn, implies that these behaviors – insofar as they are made over money – are errors to be reduced, rather

than preferences to be accommodated in policy design. Although our findings link intertemporal choice anomalies to behaviors that are responsive to time delays, we also acknowledge the potential role of random choices in this; understanding the relative importance of these two sources of error is a valuable avenue for future work.

Second, our paper suggests that models of information-processing constraints may be more promising for explaining intertemporal choice over financial flows than purely preferences-based accounts (e.g., Gabaix and Laibson 2022; Gershman and Bhui 2020; Vieider 2021b). At the same time, because these models often tie these constraints to the actual elapse of time (rooting them in imperfections in time perception, or imperfect estimates of future utils), they cannot immediately explain our evidence without re-interpretation. Our evidence suggests that the difficulty of intertemporal choice stems from something more generic than temporal misperception: the difficulty of aggregating multiple components of a problem into an optimal choice.¹² Relatedly, although the hypothesis that incomplete preferences are responsible for intertemporal anomalies is not facially consistent with our results, a slight reinterpretation may harmonize such stories with our findings: people may be unable to rank different financial flows (i.e., may have incomplete time preferences) precisely because it is cognitively difficult or costly to aggregate the components of each option into an overall valuation.

Finally, our results suggest that intertemporal choices over money may be more connected to other types of intertemporal choice than has been previously thought. Initially, the literature interpreted anomalies in the money-early-or-later paradigm as direct evidence for non-exponential time preferences, but in recent years this interpretation has been questioned: Because of the fungibility of money, impatience over monetary rewards need not directly measure true time preferences. This concern has given rise to an important literature that has focused on intertemporal choices over consumption, which are immune from this concern (e.g., Augenblick, Niederle, and Sprenger 2015; Augenblick and Rabin 2019). Our findings are supportive of the conclusion that money-early-or-later experiments do not measure preferences, but not for the reasons originally raised by the literature: intertemporal choices over money fail to identify time preferences simply because they are confounded by generic responses to complexity. This raises the possibility that our findings may also extend to intertemporal consumption decisions. After all, intertemporal consumption choices require no less complex aggregation of problem components than intertemporal monetary choice, making it plausibly subject to similar barriers to preference recovery (Carrera et al. 2022; Chakraborty et al. 2017). Examining to what degree this is true

12. Note that this approach may explain additional patterns in intertemporal choice that we do not study here. In particular, reviews such as Cohen et al. (2020) partition phenomena into those that concern time delays—such as subadditivity and hyperbolicity, as studied in this paper—and those that concern payouts or utils—gain-loss asymmetries and magnitude effects. While we focus on the former category, our conclusions provide every reason to expect that information-processing constraints and their ensuing distortions should also affect the latter. Indeed, the experiments in Gershman and Bhui (2020) suggest that magnitude effects may also be driven by cognitive noise.

will require new and different types of experiments, but seems like an important next step for the literature.

Methodological implications. One broader potential methodological implication of our results is that we should be cautious in interpreting behaviors that deviate from standard theoretical benchmarks as outgrowths of preferences. A methodological lesson (in conjunction with the recent literature on choice under risk) is that the identifying assumption that (even simple-seeming) objects of choice are rationally valued by subjects cannot be made uncritically by researchers. It is, instead, an assumption that must be *demonstrated* by ruling out the alternative possibility that valuations are instead driven by complexity-derived mistakes.

Our paper showcases several methods for testing this assumption, most of which have been previously used to test the same assumption in the context of risky choice. One of these methods—implemented through mirror tasks—is to induce preferences in a standard choice task, and examine whether those preferences are reliably expressed by the subject in behavior, or if they are instead replaced by distorting heuristics. Another method is to directly measure auxiliary evidence of boundedly rational choice. For instance, people’s doubts about the quality of their own choices (CU) or direct evidence of the use of noisy choice rules. A third is to overtly attempt to manipulate complexity to see if it intensifies the behavior under study. One contribution of our paper is to collect several of these methods together in the same investigation and to show that they produce consistent results. These methods can, in principle, also be used in other contexts.

Connections to other domains and implications for behavioral theory. Finally, an important takeaway from all of the experiments reported in this paper is that cognitively costly information processing (and the errors it induces) produces a *particular type of behavioral response*: an insufficient elasticity of decisions to variation in the main parameter of the problem (in this case, the length of the time interval). This observation may suggest deep connections between intertemporal choice anomalies and other anomalies that have similarly been identified as growing out of information-processing constraints, such as in choice under uncertainty and belief formation (e.g., Augenblick, Lazarus, and Thaler 2021; Enke and Graeber 2023; Oprea 2024).¹³ In particular, two overarching messages emerge from this recent literature.

First, when decisions involve non-trivial information processing, observed behavior is often insufficiently sensitive (“attenuated”) with respect to variation in objective problem parameters, including probabilities, deterministic frequencies, time delays, and atemporal discounting iterations. We view this generic insensitivity as a principle that potentially underlies multiple behavioral economics anomalies. If true, this would suggest that apparently distinct phenomena in behavioral economics might be parsimoniously united by models built to describe the way humans manage and respond to complexity.

13. See Prelec and Loewenstein (1991), Ebert and Prelec (2007), and Epper et al. (2024) for related discussions.

Second, behavior in various domains exhibits diminishing sensitivity away from natural boundaries, such as the present in intertemporal choice or probabilities of 0% and 100% in lottery choice. Akin to our finding of CU increasing in the time delay ([Online Appendix C](#)), psychologists have long documented that decision noise increases in the distance from boundary or reference points, producing the characteristic shape of the Weber-Fechner curve. These analogies may further suggest that diminishing sensitivity in intertemporal choice is a special case of a broader phenomenon that is ultimately driven by information-processing constraints.

We believe that these patterns may have potential implications for behavioral theory. The dominant approach in behavioral economics is to craft domain-specific models—theories of probability weighting, of hyperbolic discounting, and so on. Yet if it is true that various canonical behavioral economics phenomena are ultimately rooted in information-processing constraints, then models that formalize these constraints and the responses they induce may be able to bring together economic behaviors from across different contexts.

References

- Andreoni, James and Charles Sprenger (2012). “Risk Preferences Are Not Time Preferences.” *American Economic Review*, 102(7), 3357–3376.
- Ashraf, Nava, Dean Karlan, and Wesley Yin (2006). “Tying Odysseus to the Mast: Evidence from a Commitment Savings Product in the Philippines.” *The Quarterly Journal of Economics*, 121, 635–672.
- Augenblick, Ned, Eben Lazarus, and Michael Thaler (2021). “Overinference from Weak Signals and Underinference from Strong Signals.” arXiv preprint arXiv:2109.09871.
- Augenblick, Ned, Muriel Niederle, and Charles Sprenger (2015). “Working Over Time: Dynamic Inconsistency in Real Effort Tasks.” *The Quarterly Journal of Economics*, 130, 1067–1115.
- Augenblick, Ned and Matthew Rabin (2019). “An Experiment on Time Preference and Misprediction in Unpleasant Tasks.” *Review of Economic Studies*, 86, 941–975.
- Ba, Cuimin, J. Aislinn Bohren, and Alex Imas (2022). “Over- and Underreaction to Information.” Available at SSRN. <https://ssrn.com/abstract=4274617>
- Brownback, Andy, Alex Imas, and Michael A. Kuhn (2023). “Time Preferences and Food Choice.” NBER Working Paper No. 31726.
- Carrera, Mariana, Heather Royer, Mark Stehr, Justin Sydnor, and Dmitry Taubinsky (2022). “Who Chooses Commitment? Evidence and Welfare Implications.” *The Review of Economic Studies*, 89, 1205–1244.
- Chabris, Christopher F., David Laibson, Carrie L. Morris, Jonathon P. Schuldt, and Dmitry Taubinsky (2008). “Individual Laboratory-Measured Discount Rates Predict Field Behavior.” *Journal of Risk and Uncertainty*, 37, 237.
- Chakraborty, Anujit, Evan M. Calford, Guidon Fenig, and Yoram Halevy (2017). “External and Internal Consistency of Choices Made in Convex Time Budgets.” *Experimental Economics*, 20, 687–706.
- Cohen, Jonathan, Keith Marzilli Ericson, David Laibson, and John Myles White (2020). “Measuring Time Preferences.” *Journal of Economic Literature*, 58, 299–347.
- Dertwinkel-Kalt, Markus, Holger Gerhardt, Gerhard Riener, Frederik Schwerter, and Louis Strang (2021). “Concentration Bias in Intertemporal Choice.” *Review of Economic Studies*, 89(3), 1314–1334.
- Ebert, Jane E. J. and Drazen Prelec (2007). “The Fragility of Time: Time-Insensitivity and Valuation of the Near and Far Future.” *Management Science*, 53, 1423–1438.
- Enke, Benjamin and Thomas Graeber (2023). “Cognitive Uncertainty.” *Quarterly Journal of Economics*, 138(4), 2021–2067.

- Enke, Benjamin and Cassidy Shubatt (2023). "Quantifying Lottery Choice Complexity." NBER Working Paper No. 31677.
- Epper, Thomas, Ernst Fehr, Helga Fehr-Duda, Claus Thustrup Kreiner, David Dreyer Lassen, Søren Leth-Petersen, and Gregers Nytoft Rasmussen (2020). "Time Discounting and Wealth Inequality." *American Economic Review*, 110(4), 1177–1205.
- Epper, Thomas F. and Helga Fehr-Duda, et al. (2024). "Risk in Time: The Intertwined Nature of Risk Taking and Time Discounting." *Journal of the European Economic Association*, 22(1), 310–354.
- Ericson, Keith Marzilli and David Laibson (2019). "Intertemporal Choice." In *Handbook of Behavioral Economics: Applications and Foundations 1*. Vol. 2, edited by Bert D.. Elsevier, pp. 1–67.
- Falk, Armin, Anke Becker, Thomas Dohmen, Benjamin Enke, David Huffman, and Uwe Sunde (2018). "Global Evidence on Economic Preferences." *The Quarterly Journal of Economics*, 133, 1645–1692.
- Frydman, Cary and Lawrence J. Jin (2023). "On the Source and Instability of Probability Weighting." NBER Working Paper No. 31573.
- Fudenberg, Drew and David K. Levine (2006). "A Dual-Self Model of Impulse Control." *American Economic Review*, 96(5), 1449–1476.
- Gabaix, Xavier and David Laibson (2022). "Myopia and Discounting." NBER Working Paper No. 23254.
- Gershman, Samuel J. and Rahul Bhui (2020). "Rationally Inattentive Intertemporal Choice." *Nature Communications*, 11, 3365.
- Goda, Gopi Shah, Matthew R. Levy, Colleen Flaherty Manchester, Aaron Sojourner, and Joshua Tasoff (2015). "The Role of Time Preferences and Exponential-Growth Bias in Retirement Savings." NBER Working Paper No. 21482.
- Gul, Faruk and Wolfgang Pesendorfer (2001). "Temptation and Self-Control." *Econometrica*, 69, 1403–1435.
- Halevy, Yoram (2008). "Strotz Meets Allais: Diminishing Impatience and the Certainty Effect." *American Economic Review*, 98(3), 1145–1162.
- Imai, Taisuke, Tom A. Rutter, and Colin F. Camerer (2021). "Meta-Analysis of Present-Bias Estimation using Convex Time Budgets." *The Economic Journal*, 131, 1788–1814.
- Imas, Alex, Michael Kuhn, and Vera Mironova (2022). "Waiting to Choose: The Role of Deliberation in Intertemporal Choice." *American Economic Journal: Microeconomics*, 14(3), 414–440.
- Jackson, Matthew O. and Leet Yariv (2014). "Present Bias and Collective Dynamic Choice in the Lab." *American Economic Review*, 104(12), 4184–4204.
- Khaw, Mel Win, Ziang Li, and Michael Woodford (2021). "Cognitive Imprecision and Small-Stakes Risk Aversion." *The Review of Economic Studies*, 88, 1979–2013.
- Khaw, Mel Win, Ziang Li, and Michael Woodford (2022). "Cognitive Imprecision and Stake-Dependent Risk Attitudes." NBER Working Paper No. 30417.
- Laibson, David (1997). "Golden Eggs and Hyperbolic Discounting." *The Quarterly Journal of Economics*, 112, 443–478.
- Loewenstein, George and Drazen Prelec (1992). "Anomalies in Intertemporal Choice: Evidence and an Interpretation." *The Quarterly Journal of Economics*, 107, 573–597.
- Martinez, Seung-Keun, Stephan Meier, and Charles Sprenger (2023). "Procrastination in the Field: Evidence from Tax Filing." *Journal of the European Economic Association*, 21, 1119–1153.
- Meier, Stephan and Charles Sprenger (2010). "Present-Biased Preferences and Credit Card Borrowing." *American Economic Journal: Applied Economics*, 2(1), 193–210.
- Meier, Stephan and Charles D. Sprenger (2015). "Temporal Stability of Time Preferences." *Review of Economics and Statistics*, 97, 273–286.
- Nielsen, Kirby and John Rehbeck (2022). "When Choices Are Mistakes." *American Economic Review*, 112(7), 2237–2268.
- O'Donoghue, Ted and Matthew Rabin (1999). "Doing It Now or Later." *American Economic Review*, 89(1), 103–124.

- Oprea, Ryan (2020). “What Makes a Rule Complex?” *American Economic Review*, 110(12), 3913–3951.
- Oprea, Ryan (2024). “Decisions Under Risk Are Decisions Under Complexity.” *American Economic Review*, 114(12), 3789–3811.
- Prelec, Drazen and George Loewenstein (1991). “Decision Making Over Time and Under Uncertainty: A Common Approach.” *Management Science*, 37, 770–786.
- Read, Daniel (2001). “Is Time-Discounting Hyperbolic or Subadditive?” *Journal of Risk and Uncertainty*, 23, 5–32.
- Stango, Victor and Jonathan Zinman (2009). “Exponential Growth Bias and Household Finance.” *The Journal of Finance*, 64, 2807–2849.
- Sunde, Uwe, Thomas Dohmen, Benjamin Enke, Armin Falk, David Huffman, and Gerrit Meyerheim (2022). “Patience and Comparative Development.” *The Review of Economic Studies*, 89, 2806–2840.
- Sutter, Matthias, Martin G. Kocher, Daniela Glätzle-Rützler, and Stefan T. Trautmann (2013). “Impatience and Uncertainty: Experimental Decisions Predict Adolescents’ Field Behavior.” *American Economic Review*, 103(1), 510–531.
- Thaler, Richard (1981). “Some Empirical Evidence on Dynamic Inconsistency.” *Economics Letters*, 8, 201–207.
- Vieider, F. (2021a). “Decisions under Uncertainty as Bayesian Inference.” Working Paper No. 21/1022, Faculty of Economics and Business Administration, Ghent University.
- Vieider, Ferdinand M. (2021b). “Noisy Coding of Time and Reward Discounting.” Working Paper No. 21/036, Faculty of Economics and Business Administration, Ghent University.
- Weitzman, Martin L. (2001). “Gamma Discounting.” *American Economic Review*, 91(1), 260–271.

Supplementary Data

Supplementary data are available at [JEEA](https://journals.oup.com/jeea) online.